

## NOKOLSKI, AK, USA

WMO: 998429

Lat: 52.941N

Lon: 168.871W

Elev: 5

StdP: 101.26

Time Zone: -9.00 (NAK)

Period: 86-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -5.6       | -4.1 | N/A                           | N/A | N/A  | N/A | N/A | N/A  | 25.1                  | 0.2  | 23.3 | 2.0  | 7.1                   | 310  | 0.947 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |     |      |     |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|-----|------|-----|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%  |      | 2%  |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB  | MCDB | WB  | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k) | (l)  | (m) | (n)  | (o)                  | (p)  |
| 8             | 3.1                    | 13.9            | N/A  | 12.9 | N/A  | 12.3 | N/A  | N/A                 | N/A  | N/A | N/A  | N/A | N/A  | 7.2                  | 140  |

| Dehumidification DP/MCDB and HR |     |      |     |     |      |     |     |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-----|------|-----|-----|------|-----|-----|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |     |      | 1%  |     |      | 2%  |     |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR  | MCDB | DP  | HR  | MCDB | DP  | HR  | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b) | (c)  | (d) | (e) | (f)  | (g) | (h) | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| N/A                             | N/A | N/A  | N/A | N/A | N/A  | N/A | N/A | N/A  | N/A           | N/A  | N/A  | N/A  | N/A  | N/A  | N/A               |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 21.2              | 18.7 | 16.4 | -9.0                       | 15.7 | 2.8                | 1.6 | -11.0                                              | 16.8 | -12.6      | 17.7 | -14.2      | 18.6 | -16.2      | 19.8 |
|                   |      |      | N/A                        | N/A  | N/A                | N/A | N/A                                                | N/A  | N/A        | N/A  | N/A        | N/A  | N/A        | N/A  |
|                   |      |      | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |      | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 5.1    | 1.1   | 1.4   | 1.6   | 2.8   | 5.0   | 7.1   | 9.2   | 10.3  | 9.3   | 6.6   | 4.0   | 2.5  |
|                                                                                  | DBStd       | 3.71   | 2.63  | 2.41  | 2.18  | 1.62  | 1.37  | 1.13  | 1.22  | 1.40  | 1.81  | 1.72  | 1.98  | 2.47 |
|                                                                                  | HDD10.0     | 1827   | 277   | 241   | 259   | 217   | 154   | 86    | 31    | 14    | 33    | 106   | 180   | 232  |
|                                                                                  | HDD18.3     | 4827   | 535   | 474   | 517   | 467   | 412   | 336   | 284   | 250   | 270   | 362   | 430   | 490  |
|                                                                                  | CDD10.0     | 41     | 0     | 0     | 0     | 0     | 0     | 0     | 6     | 22    | 13    | 1     | 0     | 0    |
|                                                                                  | CDD18.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH23.3     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                                  | CDH26.7     | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                             | WSAvg       | 7.4    | 8.3   | 8.4   | 7.1   | 7.6   | 6.9   | 6.5   | 6.2   | 6.0   | 7.1   | 8.5   | 8.5   | 8.3  |
| Precipitation                                                                    | PrecAvg     | 1358   | 153   | 139   | 119   | 71    | 79    | 65    | 89    | 101   | 115   | 124   | 142   | 162  |
|                                                                                  | PrecMax     | 1707   | 266   | 201   | 251   | 113   | 133   | 102   | 137   | 149   | 169   | 213   | 277   | 277  |
|                                                                                  | PrecMin     | 1012   | 68    | 47    | 41    | 37    | 37    | 27    | 23    | 49    | 42    | 86    | 77    | 68   |
|                                                                                  | PrecStd     | 168    | 43    | 36    | 49    | 19    | 21    | 17    | 26    | 25    | 32    | 26    | 45    | 45   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 7.4   | 6.0   | 6.4   | 7.7   | 9.5   | 11.3  | 13.5  | 15.2  | 16.4  | 12.3  | 9.7   | 8.5  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | DB     | 5.7   | 5.3   | 5.7   | 6.3   | 8.2   | 10.2  | 12.5  | 13.8  | 14.7  | 10.9  | 8.5   | 7.4  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | DB     | 5.1   | 4.9   | 5.0   | 5.6   | 7.4   | 9.5   | 11.8  | 13.0  | 12.8  | 9.9   | 7.6   | 6.6  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | DB     | 4.4   | 4.4   | 4.5   | 5.1   | 6.9   | 8.9   | 11.1  | 12.6  | 11.8  | 9.1   | 6.9   | 5.9  |
|                                                                                  |             | MCWB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 2%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5%          | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 10%         | WB     | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCDB   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 3.3   | 3.2   | 3.5   | 3.2   | 3.1   | 2.8   | 2.7   | 3.1   | 3.3   | 3.1   | 3.5   | 3.3  |
|                                                                                  |             | MCDBR  | 3.0   | 3.2   | 3.8   | 3.9   | 3.8   | 3.9   | 3.6   | 3.8   | 4.2   | 3.7   | 4.0   | 3.7  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  | 5% WB       | MCDBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
|                                                                                  |             | MCWBR  | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.321  | 0.348 | 0.361 | 0.417 | 0.407 | 0.402 | 0.404 | 0.390 | 0.382 | 0.355 | 0.338 | 0.309 |      |
|                                                                                  | taud        | 2.257  | 2.255 | 2.274 | 2.115 | 2.223 | 2.323 | 2.383 | 2.425 | 2.337 | 2.302 | 2.205 | 2.245 |      |
|                                                                                  | Ebn at Noon | 641    | 741   | 818   | 809   | 842   | 850   | 842   | 834   | 789   | 722   | 599   | 578   |      |
|                                                                                  | Edh at Noon | 65     | 88    | 105   | 139   | 131   | 120   | 112   | 102   | 98    | 82    | 67    | 55    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 0.63   | 1.25  | 2.39  | 3.52  | 4.00  | 4.04  | 3.56  | 3.20  | 2.68  | 1.71  | 0.81  | 0.48  |      |
|                                                                                  | RadStd      | 0.05   | 0.08  | 0.23  | 0.34  | 0.38  | 0.35  | 0.30  | 0.23  | 0.19  | 0.11  | 0.07  | 0.05  |      |

## Historical Trends

|                       | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|-----------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                       |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| Station Only          | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| Regional (1 neighbor) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
|                       | N/A   | N/A     | +1.60  | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page